



# Locomotion and Movement

**Movement is one of the significant features of living beings.** Animals and plants exhibit a wide range of movements. The **streaming of protoplasm** in unicellular organisms like *Amoeba* is a simple form of movement; the movement of **cilia, flagella and tentacles** is shown by many organisms; and human beings can move their **limbs, jaws, eyelids, tongue**, etc.

- **Locomotion** - some movements result in a change of place or location. Such voluntary movements are called locomotion. Walking, running, climbing, flying and swimming are all forms of locomotory movements.
- Locomotory structures **need not be different** from those affecting other types of movements. For example, in *Paramecium*, cilia help in the movement of food through the cytopharynx *and* in locomotion as well; *Hydra* can use its tentacles for capturing prey *and* for locomotion; and we use our limbs for changes in body posture *and* for locomotion.

These observations suggest that movements and locomotion **cannot be studied separately**. The two may be linked by stating that **all locomotions are movements but all movements are not locomotions**.

- Methods of locomotion vary with the animal's **habitat** and the demand of the situation. However, locomotion is **generally** for the search of food, shelter, mate, suitable breeding grounds, favourable climatic conditions, or to escape from enemies/predators.

17.1

## Types of Movement

Cells of the human body exhibit **three main types of movements**, namely, **amoeboid, ciliary and muscular**.

| Type            | Where it occurs                                                          | How it works                                                                                                                                                                                                         |
|-----------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Amoeboid</b> | Specialised cells like <b>macrophages</b> and <b>leucocytes</b> in blood | Effected by <b>pseudopodia</b> formed by the streaming of protoplasm (as in <i>Amoeba</i> ); <b>cytoskeletal elements like microfilaments</b> are also involved.                                                     |
| <b>Ciliary</b>  | Most internal tubular organs lined by <b>ciliated epithelium</b>         | Coordinated cilia in the <b>trachea</b> help remove dust particles and foreign substances inhaled with air; the passage of <b>ova</b> through the female reproductive tract is also facilitated by ciliary movement. |
| <b>Muscular</b> | Limbs, jaws, tongue, etc.                                                | Uses the <b>contractile property of muscles</b> ; effectively used for locomotion and other movements by human beings and the <b>majority</b> of multicellular organisms.                                            |

① **[NOTE]** Locomotion requires a **perfect coordinated activity of the muscular, skeletal and neural systems**. This chapter covers the types of muscles, their structure, the mechanism of their contraction, and the important aspects of the skeletal system.

17.2

## Muscle

**Cilia and flagella are outgrowths of the cell membrane.** Flagellar movement helps in the **swimming of spermatozoa**, the maintenance of water current in the **canal system of sponges**, and in the locomotion of Protists like *Euglena*.

- **Muscle** - a specialised tissue of **mesodermal origin**. About **40-50 per cent** of the body weight of a human adult is contributed by muscles.
- Muscles have special properties: **excitability, contractility, extensibility and elasticity**.

Muscles have been classified using different criteria, namely **location, appearance and nature of regulation** of their activities. Based on their **location**, three types are identified: **(i) Skeletal (ii) Visceral and (iii) Cardiac**.

| Muscle type     | Location                                                           | Appearance                                     | Regulation                                                               | Chief role                                                      |
|-----------------|--------------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Skeletal</b> | Closely associated with the <b>skeletal components</b> of the body | <b>Striped / striated</b> under the microscope | Under <b>voluntary</b> control of the nervous system (voluntary muscles) | Primarily <b>locomotory actions</b> and changes of body posture |

| Muscle type     | Location                                                                           | Appearance                                                | Regulation                                                             | Chief role                                                                          |
|-----------------|------------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Visceral</b> | Inner walls of hollow visceral organs - alimentary canal, reproductive tract, etc. | <b>No striation, smooth</b> (smooth / nonstriated muscle) | <b>Not</b> under voluntary control (involuntary muscles)               | Transport of food through the digestive tract and gametes through the genital tract |
| <b>Cardiac</b>  | Muscles of the <b>heart</b> ; many cells assemble in a <b>branching pattern</b>    | <b>Striated</b>                                           | <b>Involuntary</b> - the nervous system does not control them directly | Contraction of the heart                                                            |

**Structure of a skeletal muscle.** Each organised skeletal muscle is made of a number of **muscle bundles** or **fascicles** held together by a common collagenous connective tissue layer called **fascia**. Each muscle bundle contains a number of **muscle fibres** (Figure 17.1). The bundle is also served by a **blood capillary** supply drawn in the cross-section.

- **Sarcolemma / sarcoplasm.** Each muscle fibre is lined by the plasma membrane called **sarcolemma** enclosing the **sarcoplasm**. A muscle fibre is a **syncytium** (source spelling) as the sarcoplasm contains **many nuclei**.
- The endoplasmic reticulum, i.e. the **sarcoplasmic reticulum** of the muscle fibres, is the **store house of calcium ions** ( $\text{Ca}^{++}$ ).
- A characteristic feature of the muscle fibre is a large number of parallelly arranged **filaments** in the sarcoplasm called **myofilaments** or **myofibrils**.

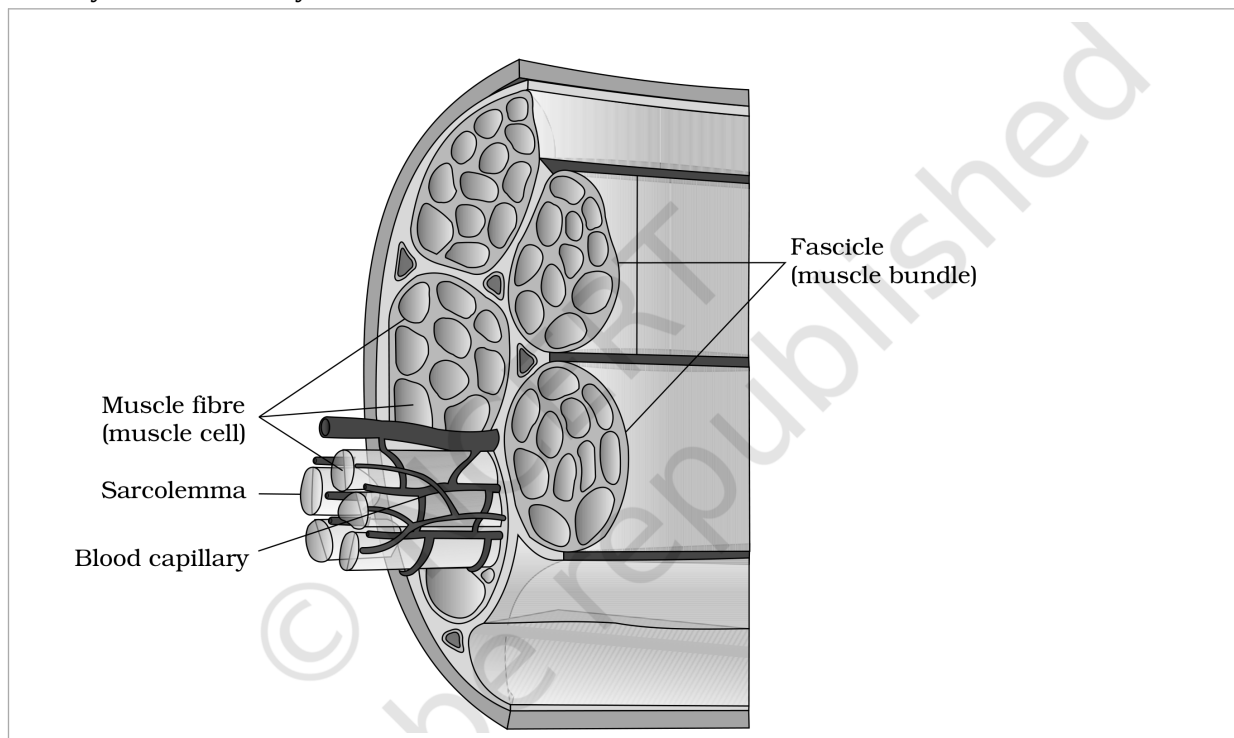


Figure 17.1 Diagrammatic cross sectional view of a muscle showing muscle bundles and muscle fibres

**Banding pattern.** Each myofibril has **alternate dark and light bands** on it. The striated appearance is due to the distribution pattern of two important proteins - **Actin and Myosin**.

| Band / line             | Also called                           | Contents                                                                                                  |
|-------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>'T' band (light)</b> | Isotropic band                        | Contains <b>actin</b> (thin filament)                                                                     |
| <b>'A' band (dark)</b>  | Anisotropic band                      | Contains <b>myosin</b> (thick filament)                                                                   |
| <b>'Z' line</b>         | Elastic fibre bisecting each 'T' band | Thin filaments are <b>firmly attached</b> to it                                                           |
| <b>'M' line</b>         | Thin fibrous membrane                 | Holds the thick filaments together in the <b>middle of the 'A' band</b>                                   |
| <b>'H' zone</b>         | Central part of the 'A' band          | The part of the thick filament <b>not overlapped</b> by thin filaments - i.e. <b>thick filaments only</b> |

- Both proteins are arranged as **rod-like structures**, parallel to each other and to the longitudinal axis of the myofibrils. **Actin filaments are thinner** than the myosin filaments, hence they are commonly called **thin** and **thick** filaments respectively.
- The '**A**' and '**I**' bands are arranged **alternately** throughout the length of the myofibrils.
- **Sarcomere** - the portion of the myofibril between **two successive 'Z' lines**. It is the **functional unit of contraction** (Figure 17.2). (The muscle **fibre** itself is the **anatomical unit** of muscle, while the sarcomere is its functional unit.)
- In a **resting state**, the edges of the thin filaments on either side of the thick filaments **partially overlap** the free ends of the thick filaments, leaving the central part of the thick filaments (the '**H**' zone) free.
- Each **sarcomere** therefore has one central '**A**' band made of **thick myosin filaments**, and **two half 'I' bands** made of **thin actin filaments** on either side of it, marked off by '**Z**' lines.

☆ **[MEMORY AID - not in NCERT]** Mnemonic for what shortens: only the '**I**' band and the '**H**' zone shrink on contraction - the '**A**' band never does.

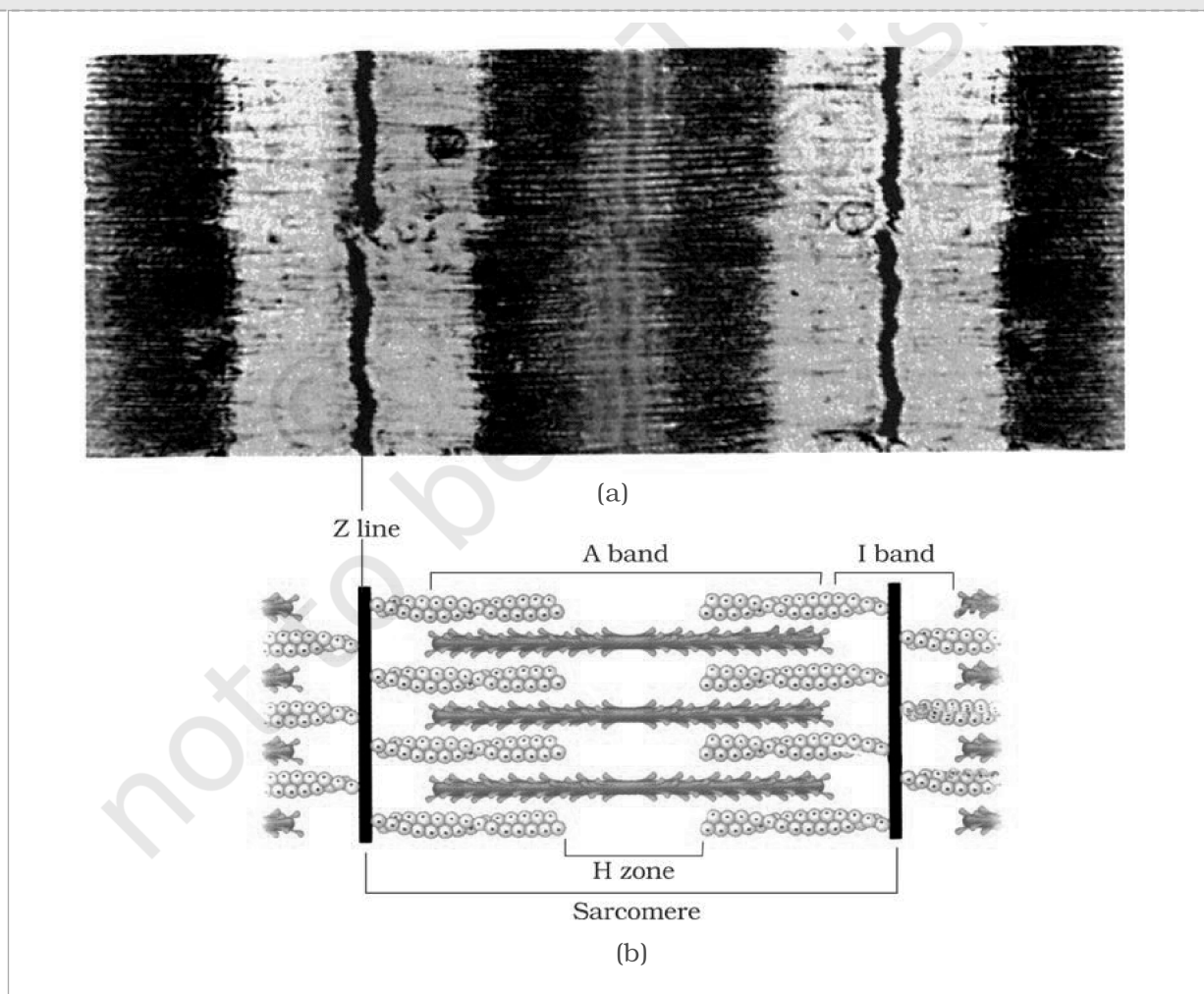


Figure 17.2 Diagrammatic representation of (a) anatomy of a muscle fibre showing a sarcomere (b) a sarcomere

17.2.1

### Structure of Contractile Proteins

**Actin (thin) filament.** Each actin filament is made of two '**F**' (**filamentous**) actins helically wound to each other. Each 'F' actin is a polymer of monomeric '**G**' (**Globular**) actins.

- Two filaments of another protein, **tropomyosin**, also run close to the 'F' actins throughout their length.
- A complex protein **Troponin** is distributed at regular intervals on the tropomyosin. In the **resting state**, a subunit of troponin **masks the active binding sites for myosin** on the actin filaments (Figure 17.3a).

**Myosin (thick) filament.** Each myosin filament is also a polymerised protein. Many monomeric proteins called **Meromyosins** (Figure 17.3b) constitute one thick filament.

| Part of a meromyosin             | Name                          | Feature                                                                                                                                                                                                                           |
|----------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Globular <b>head</b> + short arm | <b>Heavy meromyosin (HMM)</b> | Projects outwards at regular distance and angle from the surface of the filament, forming the <b>cross arm</b> ; the head is an <b>active ATPase enzyme</b> with binding sites for <b>ATP</b> and active sites for <b>actin</b> . |
| <b>Tail</b>                      | <b>Light meromyosin (LMM)</b> | The rod-like shaft of the monomer.                                                                                                                                                                                                |

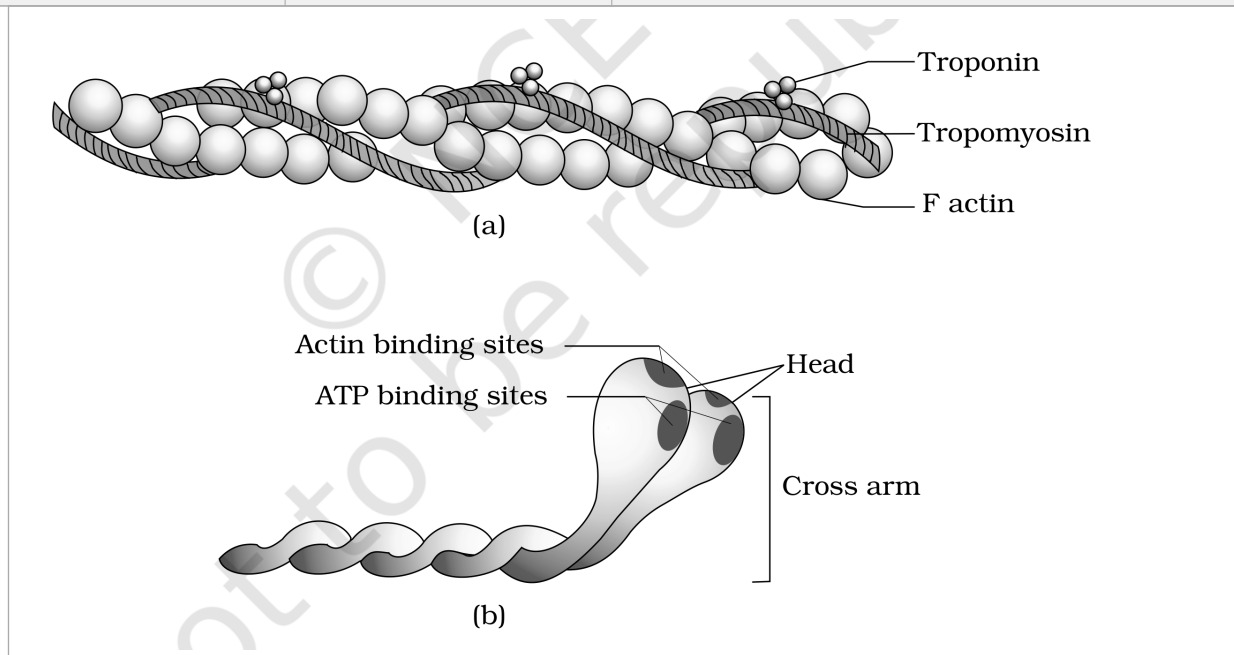


Figure 17.3 (a) An actin (thin) filament (b) Myosin monomer (Meromyosin)

17.2.2

### Mechanism of Muscle Contraction

- **Sliding filament theory** - the mechanism of muscle contraction is best explained by this theory, which states that contraction of a muscle fibre takes place by the **sliding of the thin filaments over the thick filaments**.
- **Motor unit** - a motor neuron along with the muscle fibres connected to it. The junction between a motor neuron and the sarcolemma of the muscle fibre is the **neuromuscular junction** or **motor-end plate**.

**The contraction cycle**, step by step:

- 1 Contraction is initiated by a signal sent by the **central nervous system (CNS)** via a **motor neuron**.
- 2 The neural signal reaching the neuromuscular junction releases a neurotransmitter (**Acetyl choline**), which generates an **action potential** in the sarcolemma.
- 3 The action potential spreads through the muscle fibre and causes the release of **calcium ions ( $\text{Ca}^{++}$ )** into the sarcoplasm.
- 4 The rise in  $\text{Ca}^{++}$  level leads to the binding of calcium with a subunit of **troponin** on the actin filaments, which **removes the masking** of the active sites for myosin.
- 5 Using energy from **ATP hydrolysis**, the **myosin head** binds to the exposed active sites on actin to form a **cross bridge** (Figure 17.4).
- 6 The cross bridge **pulls the attached actin filaments towards the centre of the 'A' band**; the 'Z' lines attached to these actins are also pulled inwards, **shortening the sarcomere** - i.e. contraction.
- 7 The myosin head, releasing the **ADP and  $\text{P}_i$** , goes back to its relaxed state. A **new ATP** binds and the cross bridge is **broken** (Figure 17.4).
- 8 The ATP is again hydrolysed by the myosin head and the cycle of cross-bridge formation and breakage repeats, causing further **sliding**.

The process continues till the  $\text{Ca}^{++}$  ions are **pumped back** into the sarcoplasmic cisternae, masking the actin filaments again; the 'Z' lines return to their original position - i.e. **relaxation**.

- During shortening (contraction), the '**I**' bands **get reduced**, whereas the '**A**' bands **retain their length** (Figure 17.5). The figure stages this across three states of the sarcomere: **Relaxed**, **Contracting** and **Maximally Contracted**.
- The **reaction time** of the fibres can vary in different muscles.

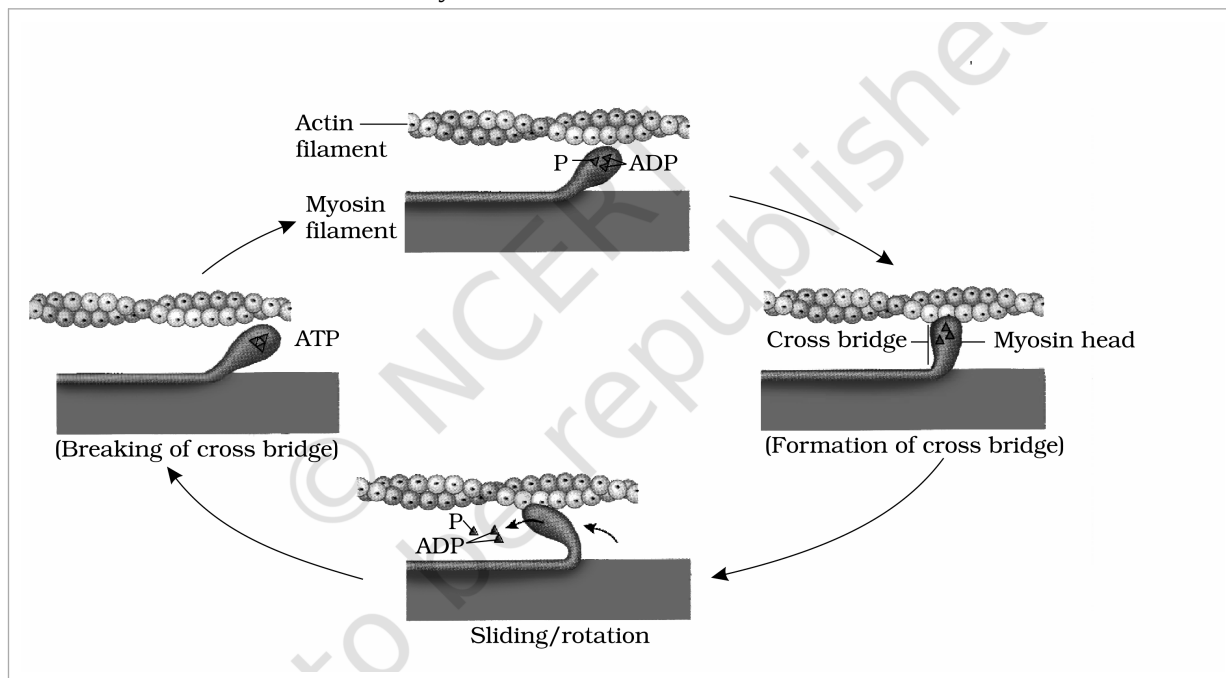


Figure 17.4 Stages in cross bridge formation, rotation of head and breaking of cross bridge

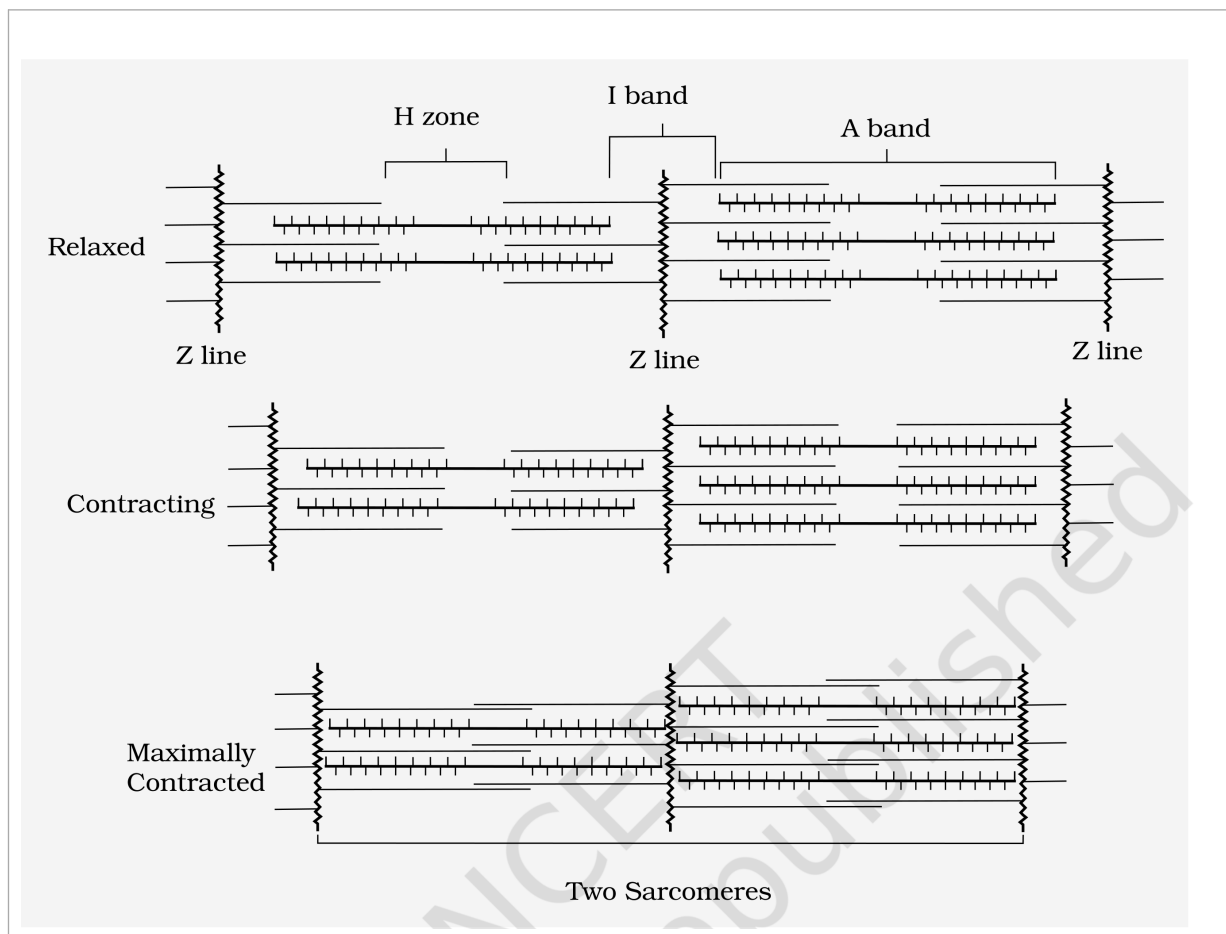


Figure 17.5 Sliding-filament theory of muscle contraction (movement of the thin filaments and the relative size of the I band and H zones)

**Fatigue.** Repeated activation of the muscles can lead to the accumulation of **lactic acid** due to the **anaerobic breakdown of glycogen** in them, causing **fatigue**.

**Red vs White fibres.** Muscles are classified as Red and White fibres based primarily on the amount of the red-coloured oxygen-storing pigment **myoglobin**.

| Feature                | Red fibres                                        | White fibres                                      |
|------------------------|---------------------------------------------------|---------------------------------------------------|
| Myoglobin              | <b>High</b> - gives a reddish appearance          | <b>Low</b> - appear pale / whitish                |
| Mitochondria           | <b>Plenty</b>                                     | <b>Few</b>                                        |
| Sarcoplasmic reticulum | Lower                                             | <b>High</b>                                       |
| Energy source          | Use stored $O_2$ for ATP - <b>aerobic muscles</b> | Depend on the <b>anaerobic</b> process for energy |

17.3

## Skeletal System

The **skeletal system** consists of a framework of **bones and a few cartilages**. This system has a significant role in the movement shown by the body - imagine chewing food without jaw bones and walking around without limb bones.

- **Bone and cartilage** are specialised **connective tissues**. Bone has a very **hard matrix** due to **calcium salts**; cartilage has a slightly **pliable matrix** due to **chondroitin salts**.
- In human beings, the skeletal system is made up of **206 bones** and a few cartilages, grouped into two principal divisions - the **axial** and the **appendicular** skeleton.

The **axial skeleton** comprises **80 bones** distributed along the main axis of the body: the **skull, vertebral column, sternum and ribs**.

**Skull** (Figure 17.6). The skull is composed of two sets of bones - **cranial and facial** - that total **22 bones**. Cranial bones are **8** in number and form the hard protective outer covering, the **cranium**, for the brain. The facial region is made up of **14** skeletal elements that form the front part of the skull.

- A single **U-shaped bone called hyoid** is present at the base of the buccal cavity.
- Each middle ear contains **three tiny bones - Malleus, Incus and Stapes**, collectively called **Ear Ossicles**.
- The skull articulates with the superior region of the vertebral column with the help of **two occipital condyles (dicondylic skull)**.

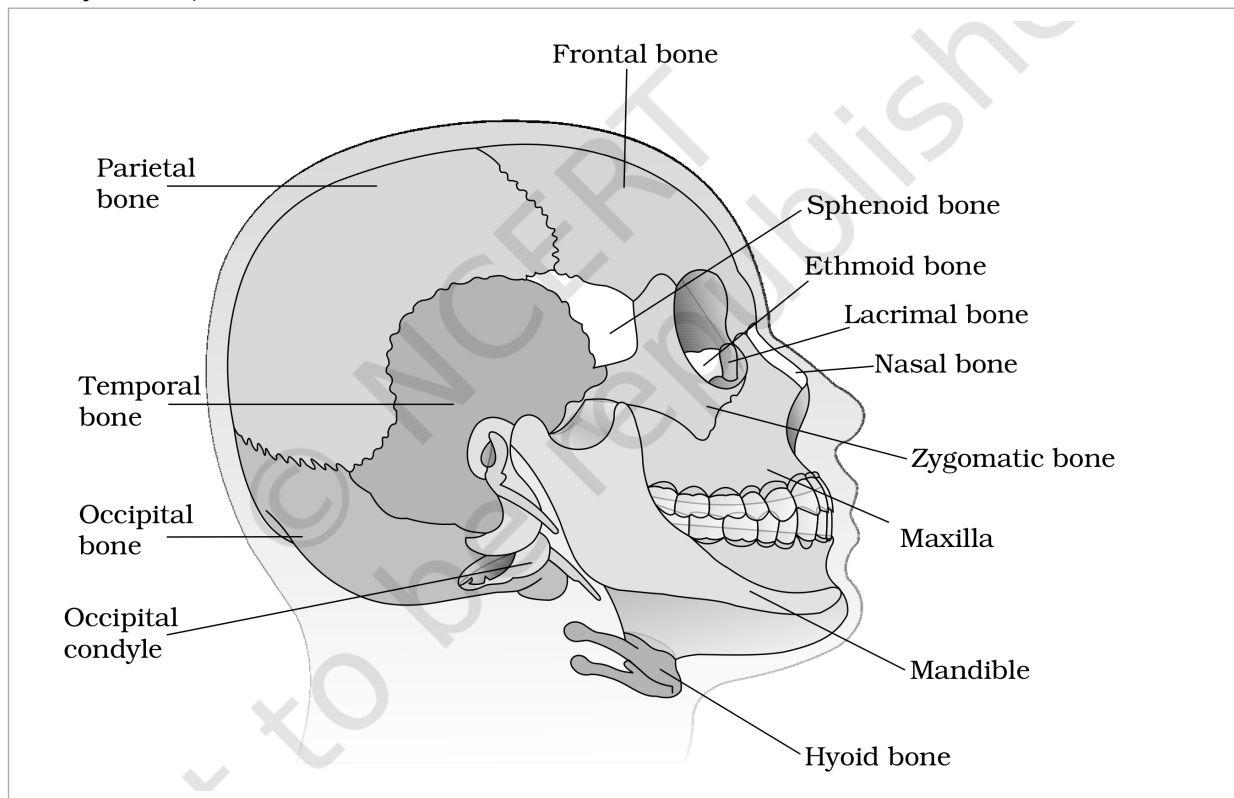


Figure 17.6 Diagrammatic view of human skull

**Vertebral column** (Figure 17.7) is formed by **26 serially arranged units called vertebrae** and is dorsally placed. It extends from the base of the skull and constitutes the main framework of the trunk.

- Each vertebra has a central hollow portion (**neural canal**) through which the **spinal cord** passes. The first vertebra is the **atlas**, and it articulates with the occipital condyles.
- The column is differentiated into **cervical (7), thoracic (12), lumbar (5), sacral (1-fused) and coccygeal (1-fused)** regions, starting from the skull. The number of **cervical vertebrae is seven in almost all mammals**, including human beings. (Adjacent vertebrae are separated by an **intervertebral disc** of cartilage, which forms the cartilaginous joint discussed in 17.4.)
- The vertebral column **protects the spinal cord, supports the head**, and serves as the point of attachment for the ribs and the musculature of the back.

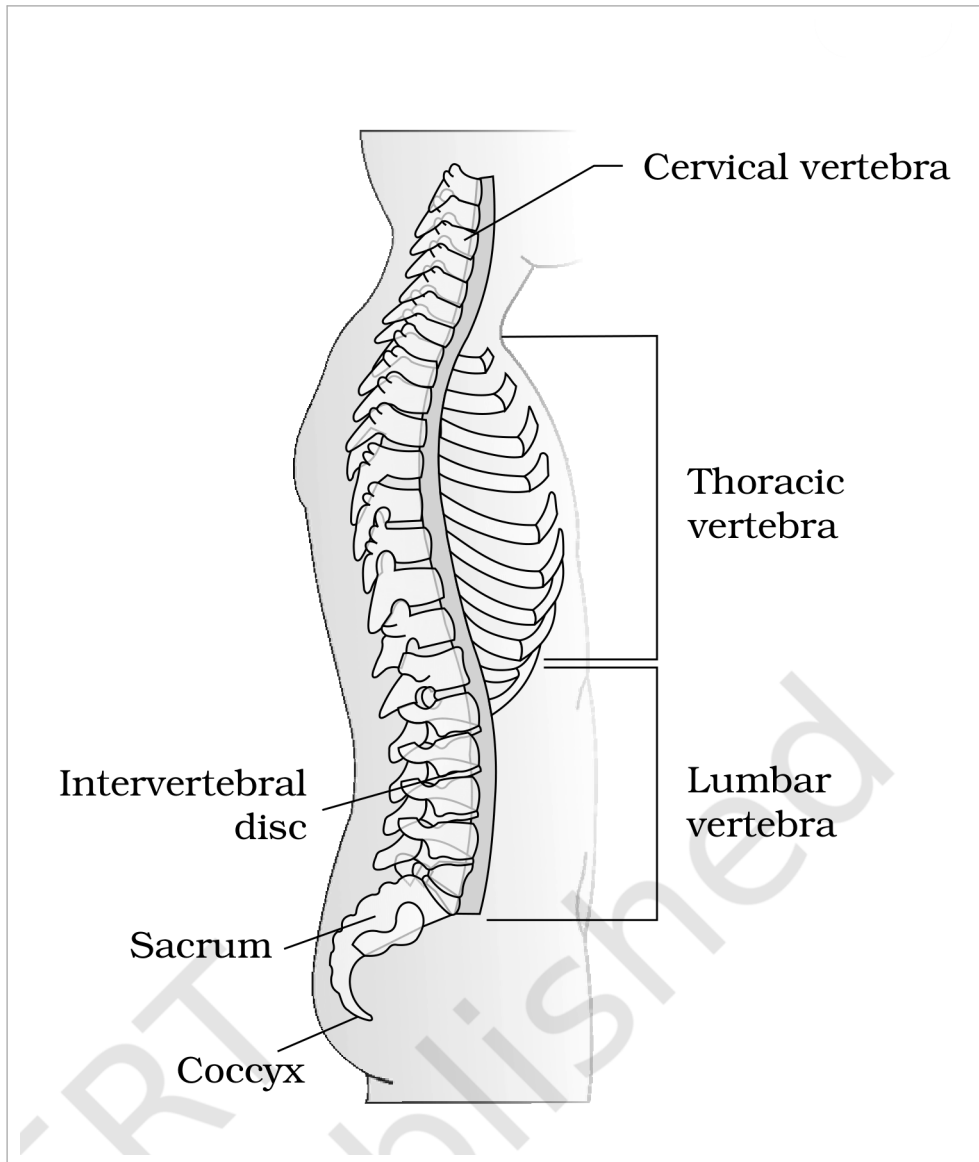


Figure 17.7 Vertebral column (right lateral view)

**Sternum and ribs.** The **sternum** is a flat bone on the ventral midline of the thorax. There are **12 pairs of ribs**. Each rib is a thin flat bone connected dorsally to the vertebral column and ventrally to the sternum; it has **two articulation surfaces** on its dorsal end and is hence called **bicephalic**.

| Rib set                              | Pairs                       | Attachment                                                                                                    |
|--------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>True ribs</b>                     | First 7 pairs               | Dorsally attached to the thoracic vertebrae; ventrally connected to the sternum with <b>hyaline cartilage</b> |
| <b>Vertebrochondral (false) ribs</b> | <b>8th, 9th, 10th</b> pairs | Do <b>not</b> articulate directly with the sternum; join the <b>seventh rib</b> with hyaline cartilage        |
| <b>Floating ribs</b>                 | Last 2 pairs (11th, 12th)   | <b>Not connected ventrally</b>                                                                                |

- Thoracic vertebrae, ribs and sternum together form the **rib cage** (Figure 17.8).

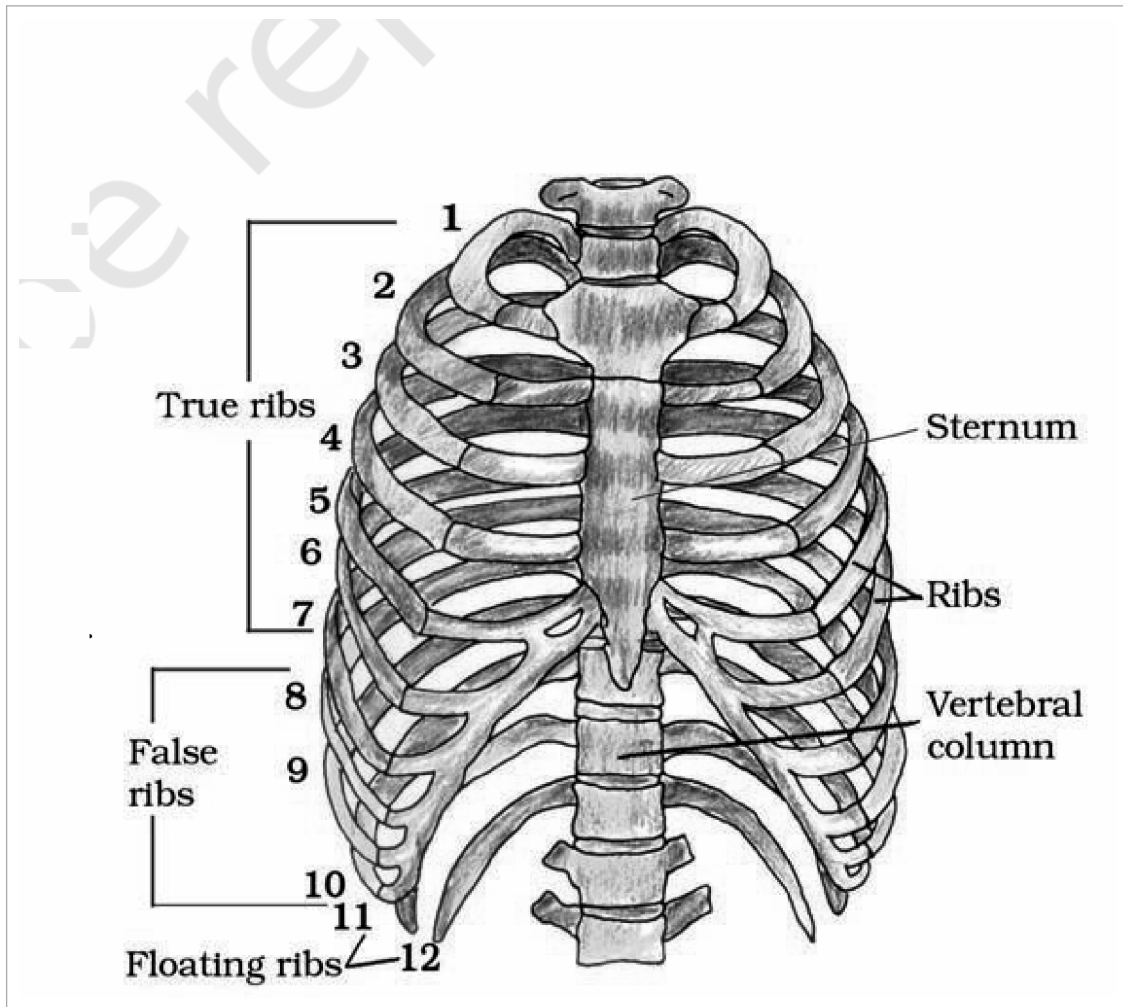


Figure 17.8 Ribs and rib cage

17.3.append

### Appendicular Skeleton

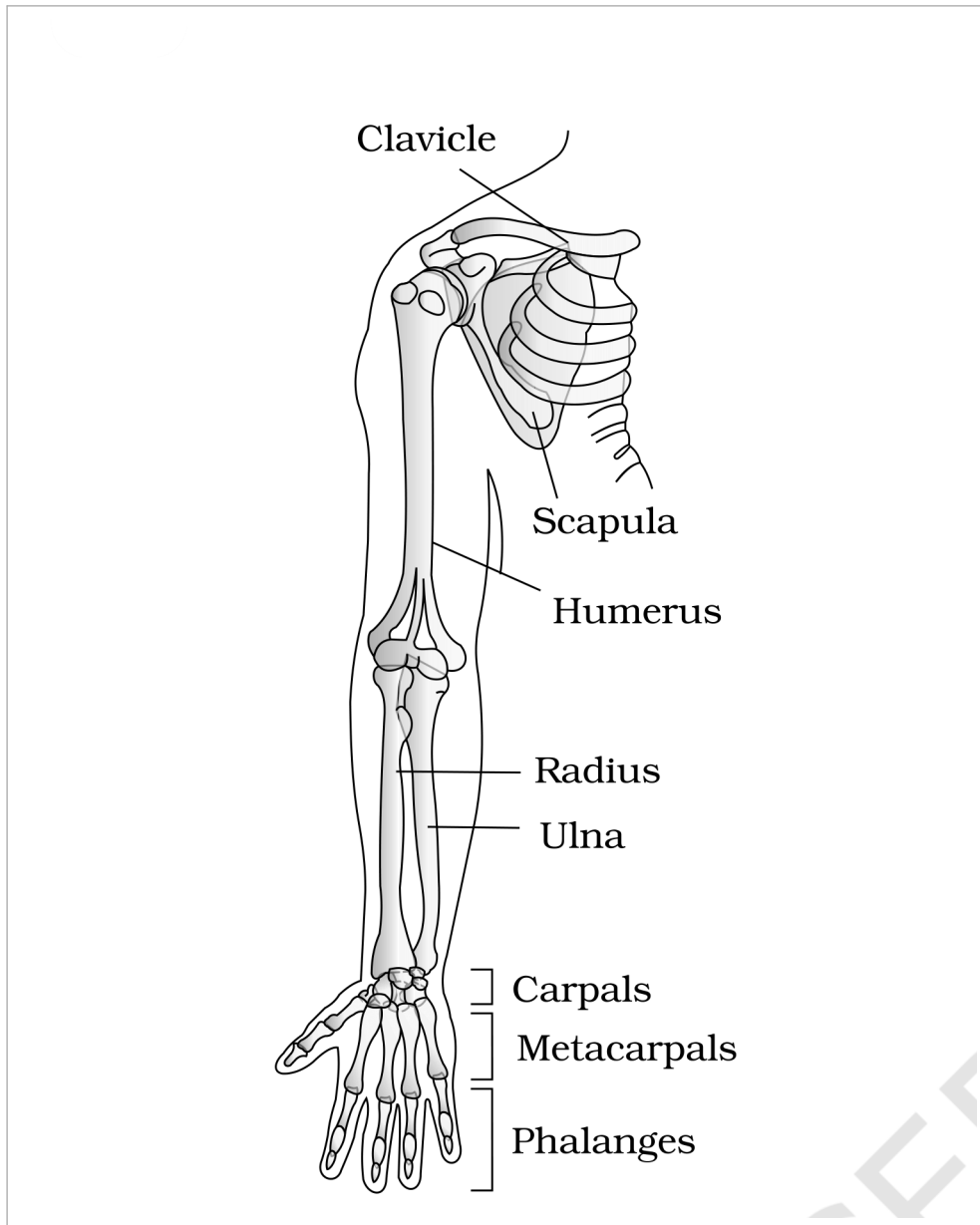
- **Appendicular skeleton** - the bones of the limbs along with their girdles. Each limb is made of **30 bones**.

| Limb                                  | Bones (with counts)                                                                                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hand / fore limb</b> (Figure 17.9) | Humerus, radius and ulna, <b>carpals</b> (wrist bones - 8), <b>metacarpals</b> (palm bones - 5) and <b>phalanges</b> (digits - 14).                                  |
| <b>Leg / hind limb</b> (Figure 17.10) | <b>Femur</b> (thigh bone - the <b>longest bone</b> ), tibia and fibula, <b>tarsals</b> (ankle bones - 7), <b>metatarsals</b> (5) and <b>phalanges</b> (digits - 14). |

- A cup-shaped bone called **patella** covers the knee ventrally (the **knee cap**).
- **Pectoral** and **Pelvic** girdle bones help in the articulation of the upper and lower limbs respectively with the axial skeleton. Each girdle is formed of two halves.

#### **Pectoral girdle** (Figure 17.9):

- Each half consists of a **clavicle** and a **scapula**.
- **Scapula** is a large **triangular flat bone** situated in the dorsal part of the thorax between the **second and the seventh ribs**. Its dorsal, flat, triangular body has a slightly elevated ridge called the **spine**, which projects as a flat, expanded process called the **acromion**; the clavicle articulates with this.
- Below the acromion is a depression called the **glenoid cavity**, which articulates with the **head of the humerus** to form the **shoulder joint**.
- Each **clavicle** is a long slender bone with two curvatures, commonly called the **collar bone**.



*Figure 17.9 Right pectoral girdle and upper arm. (frontal view)*

**Pelvic girdle** (Figure 17.10):

- Consists of **two coxal bones**. Each coxal bone is formed by the fusion of **three bones - ilium, ischium and pubis**.
- At the point of fusion is a cavity called the **acetabulum**, to which the thigh bone (femur) articulates.
- The two halves of the pelvic girdle meet ventrally to form the **pubic symphysis**, containing fibrous cartilage.

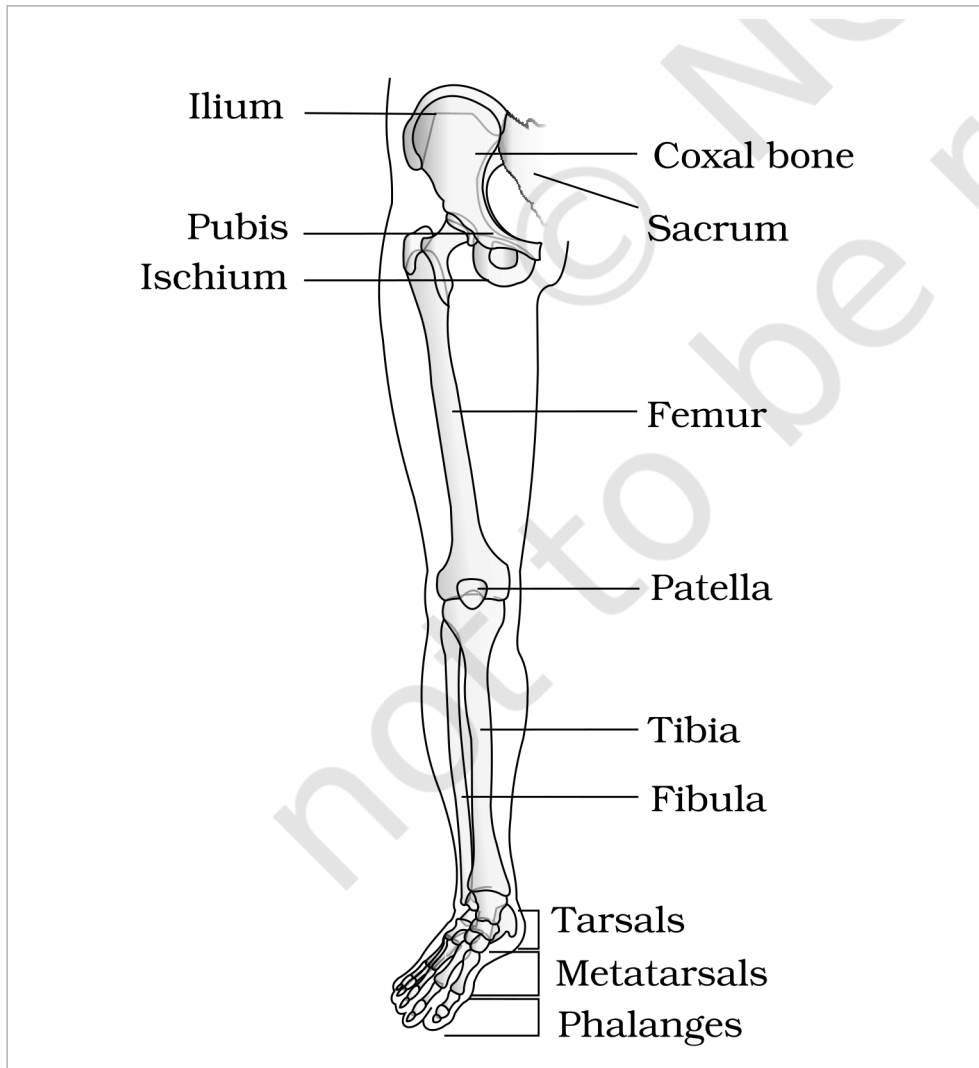


Figure 17.10 Right pelvic girdle and lower limb bones (frontal view)

☆ **[MEMORY AID - not in NCERT] Pectoral vs Pelvic girdle at a glance:** pectoral = **clavicle** + **scapula** (articulates the arm at the glenoid cavity/shoulder); pelvic = one **coxal bone** per half = **ilium** + **ischium** + **pubis** fused (articulates the thigh at the acetabulum).

17.4

## Joints

**Joints** are essential for all types of movements involving the bony parts of the body; locomotory movements are no exception.

- **Joints** - points of contact between bones, or between bones and cartilages. The force generated by the muscles is used to carry out movement through joints, where the joint acts as a **fulcrum**. The movability at these joints varies depending on different factors.

Joints have been classified into **three major structural forms**, namely, **fibrous**, **cartilaginous** and **synovial**.

| Joint type           | Movement                                                                        | Example                                                                                                             |
|----------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Fibrous</b>       | Do <b>not allow any movement</b>                                                | Flat <b>skull bones</b> fusing end-to-end via dense fibrous connective tissue as <b>sutures</b> to form the cranium |
| <b>Cartilaginous</b> | Permit <b>limited</b> movements                                                 | Joint between <b>adjacent vertebrae</b> in the vertebral column                                                     |
| <b>Synovial</b>      | Allow <b>considerable</b> movement; help in locomotion and many other movements | Characterised by a fluid-filled <b>synovial cavity</b> between the articulating surfaces of the two bones           |

**Synovial joints** are characterised by the presence of a **fluid-filled synovial cavity** between the articulating surfaces of the two bones. Such an **arrangement** allows considerable movement. Sub-types with examples:

- **Ball and socket joint** - between humerus and pectoral girdle.

- **Hinge joint** - knee joint.
- **Pivot joint** - between atlas and axis.
- **Gliding joint** - between the carpals.
- **Saddle joint** - between carpal and metacarpal of the thumb.

17.5

## Disorders of Muscular and Skeletal System

| Disorder                  | Description                                                                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Myasthenia gravis</b>  | <b>Auto immune disorder</b> affecting the <b>neuromuscular junction</b> , leading to fatigue, weakening and paralysis of skeletal muscle.                          |
| <b>Muscular dystrophy</b> | <b>Progressive degeneration</b> of skeletal muscle, mostly due to a <b>genetic disorder</b> .                                                                      |
| <b>Tetany</b>             | Rapid spasms (wild contractions) in muscle due to <b>low <math>\text{Ca}^{++}</math></b> in body fluid.                                                            |
| <b>Arthritis</b>          | <b>Inflammation of joints</b> .                                                                                                                                    |
| <b>Osteoporosis</b>       | <b>Age-related</b> disorder characterised by <b>decreased bone mass</b> and increased chances of fractures. <b>Decreased levels of estrogen</b> is a common cause. |
| <b>Gout</b>               | Inflammation of joints due to the accumulation of <b>uric acid crystals</b> .                                                                                      |

① **[NOTE]** Chapter map recap: **movement** (amoeboid / ciliary / muscular) leads to **muscle** (skeletal / visceral / cardiac), whose contractile unit is the **sarcomere**; contraction runs by the **sliding filament theory**; muscles act on the **skeletal system** (206 bones: axial + appendicular) through **joints**. Disorders can strike either the muscular side (myasthenia gravis, muscular dystrophy, tetany) or the skeletal/joint side (arthritis, osteoporosis, gout).

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## SUMMARY

**Movement is an essential feature of all living beings. Protoplasmic streaming, ciliary movements, and the movements of fins, limbs, wings, etc., are some of the forms exhibited by animals. A voluntary movement that causes the animal to change its place is locomotion.** Animals move **generally** in search of food, shelter, mate, breeding ground, better climate, or to protect themselves.

The cells of the human body exhibit **amoeboid, ciliary and muscular** movements. Locomotion and many other movements require **coordinated muscular activity**. **Three types of muscles** are present in our body: **skeletal** muscles are attached to skeletal elements and are **striated and voluntary**; **visceral** muscles, in the inner walls of visceral organs, are **nonstriated and involuntary**; **cardiac** muscles, the muscles of the heart, are **striated, branched and involuntary**. Muscles possess **excitability, contractility, extensibility and elasticity**, and contribute **40-50 per cent** of an adult's body weight.

The **muscle fibre is the anatomical unit** of muscle; each fibre holds many parallelly arranged **myofibrils**, and each myofibril many serially arranged **sarcomeres**, which are the **functional units**. Each sarcomere has a central '**A**' band of thick **myosin** filaments and **two half 'I' bands** of thin **actin** filaments on either side, marked by '**Z**' lines; the '**H**' zone is the part of the thick filament not overlapped by thin filaments. Actin and myosin are **polymerised proteins with contractility**. On a resting actin filament the active sites for myosin are **masked by troponin**. The **myosin head contains ATPase** and carries **ATP binding sites** and **active sites for actin**.

A **motor neuron** carries the signal to the muscle fibre, generating an **action potential** in it; this releases  **$\text{Ca}^{++}$**  from the **sarcoplasmic reticulum**.  **$\text{Ca}^{++}$**  activates actin, which binds the **myosin head** to form a **cross bridge**. The cross bridges **pull the actin filaments**, sliding them over the myosin filaments - i.e. **contraction**, in which the '**I**' bands **shorten while the 'A' bands keep their length**.  **$\text{Ca}^{++}$**  is then **returned to the sarcoplasmic reticulum**, actin is inactivated, the cross bridges are broken and the muscle **relaxes**. Repeated stimulation leads to **fatigue** (lactic acid from anaerobic glycogen breakdown). Muscles are classified as **Red and White fibres** based primarily on the amount of the red-coloured **myoglobin** pigment in them.

**Bones and cartilages** constitute our skeletal system (**206 bones**), divisible into **axial** and **appendicular**. **Skull, vertebral column, ribs and sternum** constitute the axial skeleton (**80 bones**); **limb bones and girdles** form the appendicular skeleton (**30 bones per limb**). **Three types of joints** are formed between bones, or between bone and cartilage - **fibrous, cartilaginous and synovial**. **Synovial joints allow considerable movement** and therefore play

a significant role in locomotion. Disorders may strike the muscular side (**myasthenia gravis, muscular dystrophy, tetany**) or the skeletal/joint side (**arthritis, osteoporosis, gout**).

E

## EXERCISES

① **[NOTE]** All ten NCERT exercises are reproduced below with worked answers. Every answer is sourced from the chapter text above. Where NCERT never states the answer outright but it follows from a definition the chapter does give (Exercise 9c, 9d), that is flagged in place.

### 1. Draw the diagram of a sarcomere of skeletal muscle showing different regions.

**Answer.** Redraw **Figure 17.2(b)** above and label: the two bounding '**Z**' lines; the central '**A**' band (dark, anisotropic, thick myosin filaments) with the '**M**' line in its middle and the '**H**' zone at its centre (thick filaments only); and the '**T**' bands (light, isotropic, thin actin filaments) on either side, each Z line bisecting an 'T' band. One sarcomere = the portion between two successive 'Z' lines = one whole 'A' band plus two half 'T' bands.

### 2. Define sliding filament theory of muscle contraction.

**Answer.** The mechanism of muscle contraction is best explained by the **sliding filament theory**, which states that **contraction of a muscle fibre takes place by the sliding of the thin filaments over the thick filaments**.

### 3. Describe the important steps in muscle contraction.

- A signal from the **CNS** travels via a **motor neuron** to the **neuromuscular junction**, releasing **Acetyl choline**, which generates an **action potential** in the sarcolemma.
- The action potential spreads through the fibre and releases  $\text{Ca}^{++}$  into the sarcoplasm; calcium binds a subunit of **troponin**, **removing the masking** of the active sites for myosin.
- Using **ATP hydrolysis**, the **myosin head** binds the exposed active sites on actin to form a **cross bridge**, which **pulls the actin filaments towards the centre of the 'A' band**; the 'Z' lines are pulled inwards and the **sarcomere shortens** ('T' bands reduce, 'A' bands keep their length).
- The head releases **ADP and  $\text{P}_i$** , a new **ATP** binds, the cross bridge is **broken**, the ATP is hydrolysed again and the cycle repeats, causing further **sliding**.
- Contraction continues till  $\text{Ca}^{++}$  is **pumped back** into the sarcoplasmic cisternae, the actin filaments are masked again and the 'Z' lines return to their original position - **relaxation**.

### 4. Write true or false. If false change the statement so that it is true.

- (a) **Actin is present in thin filament - TRUE.** The light '**T**' band contains actin, and actin filaments are thinner than myosin filaments, hence **thin** filaments.
- (b) **H-zone of striated muscle fibre represents both thick and thin filaments - FALSE.** Corrected: the '**H**' zone represents **only the thick (myosin) filaments** - it is the central part of the thick filament **not overlapped** by thin filaments.
- (c) **Human skeleton has 206 bones - TRUE.** In human beings the skeletal system is made of **206 bones** and a few cartilages.
- (d) **There are 11 pairs of ribs in man - FALSE.** Corrected: there are **12 pairs of ribs** (7 true + 3 vertebrochondral/false + 2 floating).
- (e) **Sternum is present on the ventral side of the body - TRUE.** The sternum is a flat bone on the **ventral midline of the thorax**.

### 5. Write the difference between:

#### (a) Actin and Myosin

| Feature  | Actin                                                                                                                               | Myosin                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Filament | <b>Thin</b> filament                                                                                                                | <b>Thick</b> filament                                                                                  |
| Band     | Light ' <b>T</b> ' ( <b>isotropic</b> ) band                                                                                        | Dark ' <b>A</b> ' ( <b>anisotropic</b> ) band                                                          |
| Build-up | Two ' <b>F</b> ' ( <b>filamentous</b> ) actins helically wound; each 'F' actin a polymer of ' <b>G</b> ' ( <b>globular</b> ) actins | Polymerised protein of many <b>meromyosin</b> monomers ( <b>HMM</b> head + short arm, <b>LMM</b> tail) |

| Feature             | Actin                                                                                                                      | Myosin                                                                                                                           |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Associated proteins | <b>Tropomyosin</b> along its length and <b>troponin</b> at regular intervals, which masks the myosin-binding sites at rest | None; the globular head itself is an <b>active ATPase enzyme</b> with <b>ATP binding sites</b> and <b>active sites for actin</b> |
| Role in contraction | <b>Slides over</b> the thick filament and is pulled inwards                                                                | Forms the <b>cross bridge / cross arm</b> that pulls the actin                                                                   |

### (b) Red and White muscles

- **Red fibres: high myoglobin** (reddish appearance), **plenty of mitochondria**, lower sarcoplasmic reticulum, use stored **oxygen** for ATP - **aerobic muscles**.
- **White fibres: very little myoglobin** (pale/whitish), **few mitochondria**, **high** sarcoplasmic reticulum, depend on the **anaerobic** process for energy.

### (c) Pectoral and Pelvic girdle

- **Pectoral girdle:** each half = a **clavicle** (collar bone, two curvatures) + a **scapula** (triangular flat bone between the second and seventh ribs, bearing the **spine** and **acromion**). Its **glenoid cavity** receives the **head of the humerus** to form the **shoulder joint**, articulating the **upper** limb.
- **Pelvic girdle:** two **coxal bones**, each formed by the fusion of **ilium, ischium and pubis**. The **acetabulum** at the point of fusion receives the **femur**, articulating the **lower** limb; the two halves meet ventrally at the **pubic symphysis** (fibrous cartilage).

### 6. Match Column I with Column II:

| Column I          | Column II                 | Why                                                                                   |
|-------------------|---------------------------|---------------------------------------------------------------------------------------|
| (a) Smooth muscle | <b>(iv) Involuntary</b>   | Visceral/smooth muscle is not under voluntary control of the nervous system           |
| (b) Tropomyosin   | <b>(ii) Thin filament</b> | Two tropomyosin filaments run close to the 'F' actins of the thin filament            |
| (c) Red muscle    | <b>(i) Myoglobin</b>      | Red fibres carry a high content of the red oxygen-storing pigment myoglobin           |
| (d) Skull         | <b>(iii) Sutures</b>      | Flat skull bones fuse end-to-end as fibrous joints called sutures to form the cranium |

### 7. What are the different types of movements exhibited by the cells of human body?

**Answer.** Three main types: **amoeboid** (macrophages and leucocytes, by pseudopodia and microfilaments), **ciliary** (ciliated epithelium of internal tubular organs - trachea, female reproductive tract) and **muscular** (limbs, jaws, tongue, using the contractile property of muscles).

### 8. How do you distinguish between a skeletal muscle and a cardiac muscle?

| Feature     | Skeletal muscle                                                    | Cardiac muscle                                                       |
|-------------|--------------------------------------------------------------------|----------------------------------------------------------------------|
| Location    | Closely associated with the <b>skeletal components</b> of the body | Muscles of the <b>heart</b>                                          |
| Arrangement | Bundles of parallel fibres (fascicles) held by fascia              | Many cells assemble in a <b>branching pattern</b>                    |
| Appearance  | <b>Striated</b> (striped under the microscope)                     | <b>Striated</b>                                                      |
| Regulation  | <b>Voluntary</b> - under nervous control                           | <b>Involuntary</b> - the nervous system does not control it directly |
| Chief role  | <b>Locomotory actions</b> and changes of body posture              | <b>Contraction of the heart</b>                                      |

### 9. Name the type of joint between the following:

- **(a) atlas/axis - pivot joint.**
- **(b) carpal/metacarpal of thumb - saddle joint.**
- **(c) between phalanges - hinge joint.** (Applied answer: NCERT lists the hinge joint only as the knee joint; the phalangeal joints allow the same single-plane movement.)

- **(d) femur/acetabulum - ball and socket joint.** (Applied answer: NCERT gives the humerus/pectoral-girdle joint as its ball-and-socket example and states that the thigh bone articulates into the cup-like acetabulum.)
- **(e) between cranial bones - fibrous joint** (immovable **sutures**).
- **(f) between pubic bones in the pelvic girdle - cartilaginous joint** - the **pubic symphysis**, which contains fibrous cartilage.

**10. Fill in the blank spaces:**

- **(a)** All mammals (except a few) have **seven (7)** cervical vertebrae.
- **(b)** The number of phalanges in each limb of human is **14**.
- **(c)** Thin filament of myofibril contains 2 'F' actins and two other proteins, namely **tropomyosin** and **troponin**.
- **(d)** In a muscle fibre  $\text{Ca}^{++}$  is stored in the **sarcoplasmic reticulum**.
- **(e)** **11th** and **12th** pairs of ribs are called floating ribs.
- **(f)** The human cranium is made of **8** bones.